

WEEK-3 ASSIGNMENT

Step 1: Database Setup

Create Database

```
CREATE DATABASE superstore_db;  
USE superstore_db;
```

Create Raw Table

```
CREATE TABLE superstore_raw (  
    row_id INT,  
    order_id VARCHAR(50),  
    order_date VARCHAR(20),  
    ship_date VARCHAR(20),  
    ship_mode VARCHAR(50),  
    customer_id VARCHAR(50),  
    customer_name VARCHAR(100),  
    segment VARCHAR(50),  
    country VARCHAR(50),  
    city VARCHAR(100),  
    state VARCHAR(100),  
    postal_code VARCHAR(20),  
    region VARCHAR(50),  
    product_id VARCHAR(50),  
    category VARCHAR(50),  
    sub_category VARCHAR(50),  
    product_name VARCHAR(255),  
    sales DECIMAL(10,2),  
    quantity INT,  
    discount DECIMAL(5,2),  
    profit DECIMAL(10,2)
```

);

Create Customers Table

```
CREATE TABLE customers (  
    customer_id VARCHAR(20),  
    customer_name VARCHAR(100),  
    segment VARCHAR(50),  
    country VARCHAR(50),  
    city VARCHAR(50),  
    state VARCHAR(50),  
    postal_code VARCHAR(20),  
    region VARCHAR(50)  
);
```

Create Products Table

```
CREATE TABLE products (  
    product_id VARCHAR(20),  
    category VARCHAR(50),  
    sub_category VARCHAR(50),  
    product_name VARCHAR(255)  
);
```

Create Orders Table

```
CREATE TABLE orders (  
    order_id VARCHAR(20),  
    customer_id VARCHAR(20),  
    order_date DATE,  
    ship_date DATE,  
    ship_mode VARCHAR(50),  
    sales DECIMAL(10,2),  
    quantity INT,
```

```
discount DECIMAL(10,2),  
profit DECIMAL(10,2)  
);
```

Insert data into these tables using SELECT DISTINCT.

```
INSERT INTO customers  
SELECT DISTINCT  
    customer_id,  
    customer_name,  
    segment,  
    country,  
    city,  
    state,  
    postal_code,  
    region  
FROM superstore_raw;
```

```
INSERT INTO products  
SELECT DISTINCT  
    product_id,  
    category,  
    sub_category,  
    product_name  
FROM superstore_raw;
```

```
INSERT INTO orders  
SELECT DISTINCT
```

```

order_id,
customer_id,
STR_TO_DATE(order_date, '%m/%d/%Y'),
STR_TO_DATE(ship_date, '%m/%d/%Y'),
ship_mode,
sales,
quantity,
discount,
profit
FROM superstore_raw;

```

Result Grid				
Filter Rows:		Export:		Wrap Cell Content: IA
Fetch rows:				
	product_id	category	sub_category	product_name
▶	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...
	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...
	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table
	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...
	OFF-AR-10002833	Office Supplies	Art	Newell 322
	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone voIP phone
	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...
	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables
	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black
	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine
	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner...
	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders

Result Grid								
Filter Rows:		Export:		Wrap Cell Content: IA		Fetch rows:		
	customer_id	customer_name	segment	country	city	state	postal_code	region
▶	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South
	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West
	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South
	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West
	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South
	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West
	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central
	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central
	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West
	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West
	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central
	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East
	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West
	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West
	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
	MA-17560	Matt Ahelman	Home Office	United States	Houston	Texas	77095	Central

Result Grid									
		Filter Rows:			Export:	Wrap Cell Content:	Fetch rows:		
	order_id	customer_id	order_date	ship_date	ship_mode	sales	quantity	discount	profit
▶	CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	261.96	2	0.00	41.91
	CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	731.94	3	0.00	219.58
	CA-2016-138688	DV-13045	2016-06-12	2016-06-16	Second Class	14.62	2	0.00	6.87
	US-2015-108966	SO-20335	2015-10-11	2015-10-18	Standard Class	957.58	5	0.45	-383.03
	US-2015-108966	SO-20335	2015-10-11	2015-10-18	Standard Class	22.37	2	0.20	2.52
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	48.86	7	0.00	14.17
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	7.28	4	0.00	1.97
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	907.15	6	0.20	90.72
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	18.50	3	0.20	5.78
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	114.90	5	0.00	34.47
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	1706.18	9	0.20	85.31
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	911.42	4	0.20	68.36
	CA-2017-114412	AA-10480	2017-04-15	2017-04-20	Standard Class	15.55	3	0.20	5.44
	CA-2016-161389	IM-15070	2016-12-05	2016-12-10	Standard Class	407.98	3	0.20	132.59
	US-2015-118983	HP-14815	2015-11-22	2015-11-26	Standard Class	68.81	5	0.80	-123.86
	US-2015-118983	HP-14815	2015-11-22	2015-11-26	Standard Class	2.54	3	0.80	-3.82
products 8		orders 9 ×							

Query 1: Find Orders with Sales Greater than Average Sales (Subquery)

```

SELECT *
FROM orders
WHERE sales >
(
    SELECT AVG(sales)
    FROM orders
);

```

```

SELECT COUNT(*)
FROM orders
WHERE sales >
(
    SELECT AVG(sales)
    FROM orders
);

```

Insight

- Only 4,536 out of 19,028 orders (23.8%) have sales greater than the overall average sales value.
- This indicates that a relatively small proportion of transactions contribute significantly to total revenue.

Result Grid									
		Filter Rows:			Export:	Wrap Cell Content:	Fetch rows:		
	order_id	customer_id	order_date	ship_date	ship_mode	sales	quantity	discount	profit
▶	CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	261.96	2	0.00	41.91
	CA-2016-152156	CG-12520	2016-11-08	2016-11-11	Second Class	731.94	3	0.00	219.58
	US-2015-108966	SO-20335	2015-10-11	2015-10-18	Standard Class	957.58	5	0.45	-383.03
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	907.15	6	0.20	90.72
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	1706.18	9	0.20	85.31
	CA-2014-115812	BH-11710	2014-06-09	2014-06-14	Standard Class	911.42	4	0.20	68.36
	CA-2016-161389	IM-15070	2016-12-05	2016-12-10	Standard Class	407.98	3	0.20	132.59
	CA-2014-105893	PK-19075	2014-11-11	2014-11-18	Standard Class	665.88	6	0.00	13.32
	CA-2015-106320	EB-13870	2015-09-25	2015-09-30	Standard Class	1044.63	3	0.00	240.26
	US-2015-150630	TB-21520	2015-09-17	2015-09-21	Standard Class	3083.43	7	0.50	-1665.05
	CA-2016-117590	GH-14485	2016-12-08	2016-12-10	First Class	1097.54	7	0.20	123.47
	CA-2015-117415	SN-20710	2015-12-27	2015-12-31	Standard Class	532.40	3	0.32	-46.98
	CA-2015-117415	SN-20710	2015-12-27	2015-12-31	Standard Class	371.17	4	0.20	41.76
	CA-2016-105816	JM-15265	2016-12-11	2016-12-17	Standard Class	1029.95	5	0.00	298.69
	CA-2016-111682	TB-21055	2016-06-17	2016-06-18	First Class	319.41	5	0.10	7.10
	CA-2014-106376	RS-11590	2014-12-05	2014-12-10	Standard Class	1113.02	8	0.20	111.30

Query 2: Find Highest Sales Order for Each Customer (Subquery)

SELECT

c.customer_id,

c.customer_name,

c.country,

c.state,

c.city,

o.order_id,

o.sales AS highest_order_per_customer

FROM customers c

JOIN orders o

ON c.customer_id = o.customer_id

```

WHERE (o.customer_id, o.sales) IN
(
    SELECT customer_id,
           MAX(sales)
    FROM orders
    GROUP BY customer_id
)
ORDER BY highest_order_per_customer DESC;

```

Insight

- This query identifies the highest-value purchase made by each customer.
- Sean Miller recorded the highest individual order value of **22,638.48**.
- Understanding such transactions helps identify premium customers and major revenue contributors.

customer_id	customer_name	country	state	city	order_id	highest_order_per_customer
SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Florida	Jacksonville	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Florida	Jacksonville	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Indiana	Lafayette	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	Indiana	Lafayette	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48
SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48

Query 3: Calculate Total Sales per Customer (CTE)

```

WITH total_sales AS (
    SELECT
        customer_id,

```

```

        SUM(sales) AS total_sale
    FROM orders
    GROUP BY customer_id
)
SELECT
    c.customer_id,
    c.customer_name,
    c.city,
    c.state,
    c.country,
    t.total_sale
FROM customers c
JOIN total_sales t
ON c.customer_id = t.customer_id
ORDER BY total_sale DESC;

```

Insight

Top revenue-generating customers:

Customer Name Total Sales

Sean Miller	25,043.07
Tamara Chand	19,017.85
Raymond Buch	15,117.35

These customers contribute significantly to business revenue and should be prioritized for retention strategies.

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		Fetch rows:
	customer_id	customer_name	city	state	country	total_sale
▶	SM-20320	Sean Miller	Los Angeles	California	United States	50086.14
	SM-20320	Sean Miller	Allentown	Pennsylvania	United States	50086.14
	SM-20320	Sean Miller	Monroe	North Carolina	United States	50086.14
	SM-20320	Sean Miller	Lafayette	Indiana	United States	50086.14
	SM-20320	Sean Miller	Jacksonville	Florida	United States	50086.14
	SM-20320	Sean Miller	Monroe	North Carolina	United States	50086.14
	SM-20320	Sean Miller	Lafayette	Indiana	United States	50086.14
	SM-20320	Sean Miller	Jacksonville	Florida	United States	50086.14
	SM-20320	Sean Miller	Los Angeles	California	United States	50086.14
	SM-20320	Sean Miller	Allentown	Pennsylvania	United States	50086.14
	TC-20980	Tamara Chand	Lafayette	Indiana	United States	38035.70
	TC-20980	Tamara Chand	Long Beach	New York	United States	38035.70
	TC-20980	Tamara Chand	Decatur	Alabama	United States	38035.70
	TC-20980	Tamara Chand	Seattle	Washington	United States	38035.70
	TC-20980	Tamara Chand	Houston	Texas	United States	38035.70
	TC-20980	Tamara Chand	Lafayette	Indiana	United States	38035.70

Additional Analysis: Average Sales per Customer

WITH avg_sales AS (

SELECT

customer_id,

AVG(sales) AS avg_sale

FROM orders

GROUP BY customer_id

)

SELECT

c.customer_id,

c.customer_name,

c.city,

c.state,

c.country,

a.avg_sale

FROM customers c

JOIN avg_sales a

ON c.customer_id = a.customer_id

ORDER BY avg_sale DESC;

Insight

- Mitch Willingham recorded the highest average order value at **1,751.29**.
- Customers with high average sales represent premium buyers with strong purchasing power.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

	customer_id	customer_name	city	state	country	avg_sale
▶	MW-18235	Mitch Willingham	San Francisco	California	United States	1751.293333
	MW-18235	Mitch Willingham	Virginia Beach	Virginia	United States	1751.293333
	MW-18235	Mitch Willingham	San Francisco	California	United States	1751.293333
	MW-18235	Mitch Willingham	Virginia Beach	Virginia	United States	1751.293333
	TC-20980	Tamara Chand	Lafayette	Indiana	United States	1728.895455
	TC-20980	Tamara Chand	Long Beach	New York	United States	1728.895455
	TC-20980	Tamara Chand	Decatur	Alabama	United States	1728.895455
	TC-20980	Tamara Chand	Seattle	Washington	United States	1728.895455
	TC-20980	Tamara Chand	Houston	Texas	United States	1728.895455
	TC-20980	Tamara Chand	Lafayette	Indiana	United States	1728.895455
	TC-20980	Tamara Chand	Long Beach	New York	United States	1728.895455
	TC-20980	Tamara Chand	Decatur	Alabama	United States	1728.895455
	TC-20980	Tamara Chand	Seattle	Washington	United States	1728.895455
	TC-20980	Tamara Chand	Houston	Texas	United States	1728.895455
	SM-20320	Sean Miller	Monroe	North Caro...	United States	1669.538000
	SM-20320	Sean Miller	Lafayette	Indiana	United States	1669.538000

Additional Analysis: Customer Purchase Frequency

WITH freq_of_order_per_cust AS (

SELECT

customer_id,

COUNT(order_id) AS frequency

FROM orders

GROUP BY customer_id

)

SELECT

c.customer_id,

c.customer_name,

f.frequency AS no_of_orders

FROM customers c

```
JOIN freq_of_order_per_cust f
ON c.customer_id = f.customer_id
ORDER BY no_of_orders DESC;
```

Insight

- Matt Abelman and William Brown each placed 33 orders.
- High-frequency customers often indicate strong customer loyalty and repeat purchasing behavior.

Result Grid			
Filter Rows:			
	customer_id	customer_name	No_of_orders
▶	WB-21850	William Brown	33
	WB-21850	William Brown	33
	MA-17560	Matt Abelman	33
	WB-21850	William Brown	33
	MA-17560	Matt Abelman	33
	WB-21850	William Brown	33
	MA-17560	Matt Abelman	33
	WB-21850	William Brown	33
	WB-21850	William Brown	33
	MA-17560	Matt Abelman	33
	MA-17560	Matt Abelman	33

Query 4: Find Customers Whose Total Sales Are Above Average (CTE + Subquery)

```
WITH total_sales AS (
  SELECT
    customer_id,
    SUM(sales) AS total_sale
  FROM orders
  GROUP BY customer_id
)
SELECT
  c.customer_id,
  c.customer_name,
```

```

c.city,
c.state,
c.country,
ts.total_sale
FROM customers c
JOIN total_sales ts
ON c.customer_id = ts.customer_id
WHERE ts.total_sale >
(
    SELECT AVG(total_sale)
    FROM total_sales
);

```

Insight

- Customers above the average sales threshold represent the highest-value customer segment.
- These customers contribute disproportionately to overall revenue and are ideal candidates for loyalty and retention programs.

Result Grid						
Filter Rows:						
Export:  Wrap Cell Content:  Fetch rows: 						
	customer_id	customer_name	city	state	country	total_sale
▶	BH-11710	Brosina Hoffman	Los Angeles	California	United States	6255.34
	IM-15070	Irene Maddox	Seattle	Washington	United States	4930.49
	PK-19075	Pete Kriz	Madison	Wisconsin	United States	8158.65
	TB-21520	Tracy Blumstein	Philadelphia	Pennsylvania	United States	4730.62
	MA-17560	Matt Abelman	Houston	Texas	United States	4280.62
	SN-20710	Steve Nguyen	Houston	Texas	United States	3127.97
	LC-16930	Linda Cazamias	Naperville	Illinois	United States	4481.16
	RA-19885	Ruben Ausman	Los Angeles	California	United States	3546.23
	ES-14080	Erin Smith	Melbourne	Florida	United States	4648.52
	KM-16720	Kunst Miller	Los Angeles	California	United States	4909.47
	KD-16270	Karen Daniels	Springfield	Virginia	United States	8282.36

Query 5: Rank Customers Based on Total Sales (Window Function)

```

SELECT
    c.customer_id,
    c.customer_name,
    SUM(o.sales) AS total_sale,
    RANK() OVER (ORDER BY SUM(o.sales) DESC) AS rnk
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name;

```

Insight

Customer ranking provides a clear picture of revenue contribution and helps businesses identify their most valuable customers.

Result Grid				
Filter Rows:				
	customer_id	customer_name	total_sale	rnk
▶	SE-20110	Sanjit Engle	268607.68	1
	CL-12565	Clay Ludtke	261133.44	2
	SC-20095	Sanjit Chand	254562.12	3
	KL-16645	Ken Lonsdale	253294.56	4
	SM-20320	Sean Miller	250430.70	5
	GT-14710	Greg Tran	236402.40	6
	AB-10105	Adrian Barton	229689.76	7
	SV-20365	Seth Vernon	228412.80	8
	PK-19075	Pete Kriz	195807.60	9
	EH-13765	Edward Hooks	192709.40	10
	ZC-21910	Zuschuss Carroll	192507.12	11

Query 6: Assign Row Numbers to Orders within Each Customer

```

SELECT
    c.customer_id,
    c.customer_name,
    o.order_id,
    ROW_NUMBER() OVER (
        PARTITION BY c.customer_id
        ORDER BY o.order_id
    ) AS rn

```

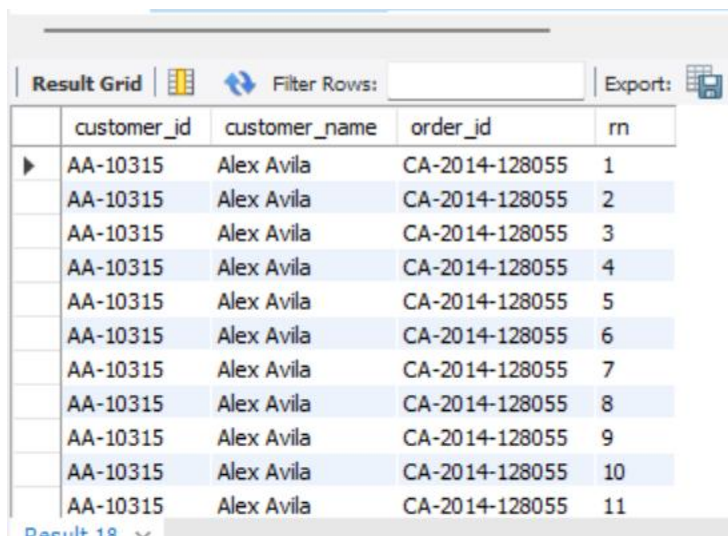
```

FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id;

```

Insight

This query helps track the sequence of purchases made by each customer and supports customer journey analysis.



The screenshot shows a database query result grid with the following columns: customer_id, customer_name, order_id, and rn. The data is filtered for customer AA-10315, showing 11 orders with order IDs ranging from CA-2014-128055. The 'rn' column indicates the sequence of purchases from 1 to 11.

customer_id	customer_name	order_id	rn
AA-10315	Alex Avila	CA-2014-128055	1
AA-10315	Alex Avila	CA-2014-128055	2
AA-10315	Alex Avila	CA-2014-128055	3
AA-10315	Alex Avila	CA-2014-128055	4
AA-10315	Alex Avila	CA-2014-128055	5
AA-10315	Alex Avila	CA-2014-128055	6
AA-10315	Alex Avila	CA-2014-128055	7
AA-10315	Alex Avila	CA-2014-128055	8
AA-10315	Alex Avila	CA-2014-128055	9
AA-10315	Alex Avila	CA-2014-128055	10
AA-10315	Alex Avila	CA-2014-128055	11

Query 7: Display Top 3 Customers Based on Total Sales

```

WITH ranked_customers AS (
    SELECT
        c.customer_id,
        c.customer_name,
        SUM(o.sales) AS total_sale,
        RANK() OVER (ORDER BY SUM(o.sales) DESC) AS rnk
    FROM customers c
    JOIN orders o
    ON c.customer_id = o.customer_id
    GROUP BY c.customer_id, c.customer_name
)
SELECT *

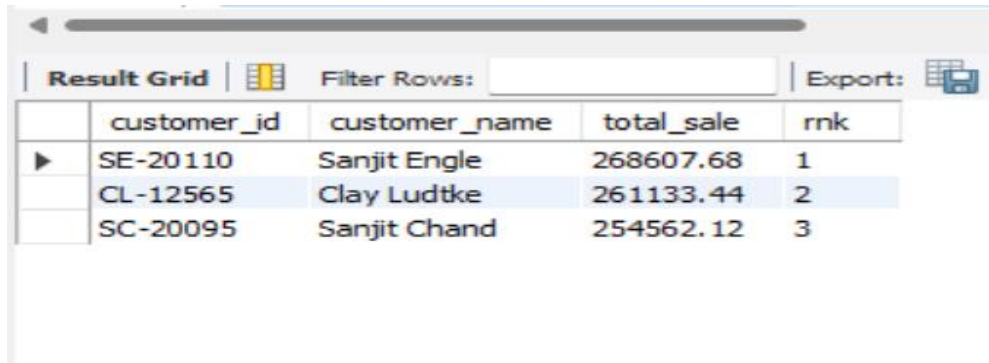
```

```
FROM ranked_customers
```

```
WHERE rnk <= 3;
```

Insight

These customers represent the highest revenue-generating segment in the dataset and contribute significantly to overall sales performance.



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with 5 columns: an empty column, 'customer_id', 'customer_name', 'total_sale', and 'rnk'. There are three rows of data, with the second row highlighted in blue. Above the table, there is a 'Filter Rows:' input field and an 'Export:' button with a download icon.

	customer_id	customer_name	total_sale	rnk
▶	SE-20110	Sanjit Engle	268607.68	1
	CL-12565	Clay Ludtke	261133.44	2
	SC-20095	Sanjit Chand	254562.12	3

Step 3: Final Combined Query

Customer Name, Total Sales, and Rank

```
WITH ranked_customers AS (  
  SELECT  
    c.customer_id,  
    c.customer_name,  
    SUM(o.sales) AS total_sale,  
    RANK() OVER (ORDER BY SUM(o.sales) DESC) AS rnk  
  FROM customers c  
  JOIN orders o  
  ON c.customer_id = o.customer_id  
  GROUP BY c.customer_id, c.customer_name  
)  
SELECT  
  customer_id,  
  customer_name,
```

```

total_sale,
rnk
FROM ranked_customers;

```

Result Grid					Filter Rows:	Export:
	customer_id	customer_name	total_sale	rnk		
▶	SE-20110	Sanjit Engle	268607.68	1		
	CL-12565	Clay Ludtke	261133.44	2		
	SC-20095	Sanjit Chand	254562.12	3		
	KL-16645	Ken Lonsdale	253294.56	4		
	SM-20320	Sean Miller	250430.70	5		
	GT-14710	Greg Tran	236402.40	6		
	AB-10105	Adrian Barton	229689.76	7		
	SV-20365	Seth Vernon	228412.80	8		
	PK-19075	Pete Kriz	195807.60	9		
	EH-13765	Edward Hooks	192709.40	10		
	ZC-21910	Zuschuss Carroll	192507.12	11		

Mini Project: Customer Sales Insights

Q1. Who are the top 5 customers?

```

with rnked_cust as (select c.customer_id,
c.customer_name,
sum(o.sales) as total_sale,
rank() over(order by sum(o.sales)DESC) as rnk
from customers c
join orders o
on c.customer_id=o.customer_id
group by c.customer_id, customer_name)
select customer_id,

```

customer_name,

total_sale,

rnk

from rnked_cust

where rnk<6;

Ans:

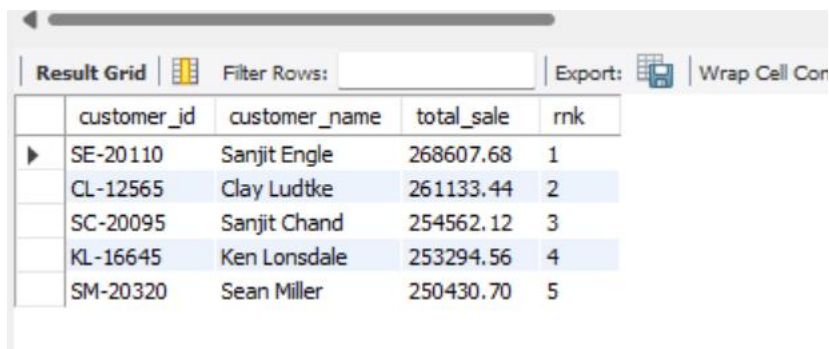
-- 1. Sanjit Engle

-- 2. Clay Ludtke

-- 3. Sanjit Chand

-- 4. Ken Lonsdale

-- 5. Sean Miller



The screenshot shows a database interface with a 'Result Grid' tab. It displays a table with 5 columns: customer_id, customer_name, total_sale, and rnk. The data is sorted by total_sale in descending order. The top 5 customers are listed, with their respective IDs and ranks.

	customer_id	customer_name	total_sale	rnk
▶	SE-20110	Sanjit Engle	268607.68	1
	CL-12565	Clay Ludtke	261133.44	2
	SC-20095	Sanjit Chand	254562.12	3
	KL-16645	Ken Lonsdale	253294.56	4
	SM-20320	Sean Miller	250430.70	5

Q2. Who are the bottom 5 customers?

with rnked_cust as (select c.customer_id,

c.customer_name,

```
sum(o.sales) as total_sale,

rank() over(order by sum(o.sales)ASC) as rnk

from customers c

join orders o

on c.customer_id=o.customer_id

group by c.customer_id, customer_name)

select customer_id,

customer_name,

total_sale,

rnk

from rnked_cust

where rnk<6;

-- 1. Lela Donovan

-- 2. Thais Sissman

-- 3. Mitch Gastineau

-- 4. Sung Chung

-- 5. Carl Jackson
```

Result Grid				
Filter Rows:		Export:		
	customer_id	customer_name	total_sale	rnk
▶	LD-16855	Lela Donovan	10.60	1
	TS-21085	Thais Sissman	19.36	2
	MG-18205	Mitch Gastineau	24.64	3
	SC-20845	Sung Chung	28.22	4
	CJ-11875	Carl Jackson	33.04	5

Q3. Which customers made only one order?

SELECT

c.customer_id,

c.customer_name,

COUNT(o.order_id) AS total_orders

FROM customers c

JOIN orders o ON c.customer_id = o.customer_id

GROUP BY c.customer_id, c.customer_name

HAVING COUNT(o.order_id) = 1;

-- No customer had only one order.

Q4. Which customers have above-average sales?

with total_sales as(

```
select customer_id,

SUM(sales) as total_sale

from orders

group by customer_id

)

select

c.customer_id,

c.customer_name,

c.city,

c.state,

c.country,

ts.total_sale as total_sales_above_avg

from customers c

JOIN total_sales ts

ON c.customer_id=ts.customer_id

where ts.total_sale>(select avg(total_sale) from total_sales);

-- Claire Gute, Darrin Van Huff, Darrin Van Huff, Brosina Hoffman,

-- Andrew Allen, Irene Maddox and 4713 others have sales above average.
```

Result Grid						
Filter Rows:						
Export:						
Wrap Cell Content:						
Fetch rows:						
	customer_id	customer_name	city	state	country	total_sale
▶	BH-11710	Brosina Hoffman	Los Angeles	California	United States	6255.34
	IM-15070	Irene Maddox	Seattle	Washington	United States	4930.49
	PK-19075	Pete Kriz	Madison	Wisconsin	United States	8158.65
	TB-21520	Tracy Blumstein	Philadelphia	Pennsylvania	United States	4730.62
	MA-17560	Matt Abelman	Houston	Texas	United States	4280.62
	SN-20710	Steve Nguyen	Houston	Texas	United States	3127.97
	LC-16930	Linda Cazamias	Naperville	Illinois	United States	4481.16
	RA-19885	Ruben Ausman	Los Angeles	California	United States	3546.23
	ES-14080	Erin Smith	Melbourne	Florida	United States	4648.52
	KM-16720	Kunst Miller	Los Angeles	California	United States	4909.47
	KD-16270	Karen Daniels	Springfield	Virginia	United States	8282.36

Q5. What is the highest order value per customer?

select

c.customer_id,

c.customer_name,

c.country,

c.state,

c.city,

o.order_id,

o.sales as highest_order_per_customer

from customers c

JOIN orders o

ON c.customer_id=o.customer_id

```

WHERE (o.customer_id, o.sales) in

(select customer_id,

Max(sales)

from orders

group by customer_id) ORDER BY highest_order_per_customer DESC;

```

Result Grid							
		Filter Rows:		Export:	Wrap Cell Content:	Fetch rows:	
	customer_id	customer_name	country	state	city	order_id	highest_order_per_customer
▶	SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Pennsylvania	Allentown	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	California	Los Angeles	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Florida	Jacksonville	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Florida	Jacksonville	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Indiana	Lafayette	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	Indiana	Lafayette	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48
	SM-20320	Sean Miller	United States	North Carolina	Monroe	CA-2014-145317	22638.48

Brief Insights

1. Only 23.8% of orders exceed the average sales value, indicating that a small proportion of transactions generate a large share of revenue.
2. Sean Miller recorded the highest individual order value (22,638.48), demonstrating the impact of high-value purchases on business performance.
3. Sean Miller, Tamara Chand, and Raymond Buch emerged as leading contributors in terms of total customer revenue.
4. Mitch Willingham achieved the highest average order value (1,751.29), identifying him as a premium customer with strong purchasing power.
5. Matt Abelman and William Brown placed the highest number of orders (33 each), indicating strong customer loyalty and repeat purchasing behavior.

6. Sanjit Engle, Clay Ludtke, and Sanjit Chand were the top three customers based on total sales.
7. The analysis demonstrates how SQL Subqueries, CTEs, and Window Functions can be used to uncover customer insights and support data-driven decision-making.